

Online Supplement to Leverage Direction Matters: Cross-Asset Evidence on GARCH Model Selection and Volatility Targeting

Yi-Hao Lai

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Online Supplementary Material

This supplement collects auxiliary null-result tables, commodity extensions, and cross-market evidence that support the main paper but are not required for the submission-length version.

Table 1: Window Size Robustness: GJR-GARCH QLIKE for SPY

OOS Period	$w = 504$	$w = 1000$	$w = 2000$	$w = 3000$	$w = 5000$
2020–2021	− 8.051	−8.027	−8.006	−8.015	−8.003
2023–2024	−8.671	−8.660	−8.652	−8.663	− 8.682
2025–2026	−8.429	−8.444	−8.438	−8.433	− 8.457

Table 2: Hybrid VT vs. Alternative Strategies (2014–2026)

Strategy	Sharpe	MaxDD	Δ vs. Hybrid
Hybrid VT	0.99	−11.4%	—
RV20 VT	0.83	−13.8%	−0.16
GARCH VT	0.82	−13.5%	−0.17
EWMA VT	0.79	−13.4%	−0.20
Buy & Hold	0.75	−33.7%	−0.24

Table 3: Diversification Amplification: ETF vs. Constituent Stock γ

Index	ETF γ	Avg Stock γ	Ratio	t -stat
SPY (50 stocks)	0.211	0.076 [†]	2.8×	−10.53***
EEM (3 stocks)	0.138	0.041	3.3×	—
EWJ (10 stocks)	0.087	0.127	0.7×	2.09
EWG (3 stocks)	0.114	0.131	0.9×	—

Notes: t -stat tests $H_0: \hat{\gamma}_{\text{ETF}} = \bar{\hat{\gamma}}_{\text{stock}}$ (Welch). *** $p < 0.001$. [†]Constituent $\bar{\hat{\gamma}}$ for SPY was originally estimated from N=50 index members (not individually verifiable via public API). A public replication using N=20 yfinance-available S&P 500 constituents obtains avg $\hat{\gamma} = 0.094$; the updated t -statistic (−10.53) is reported. Amplification direction ($\hat{\gamma}_{\text{ETF}} > \hat{\gamma}_{\text{stock}}$, $p < 0.001$) is preserved under both N=50 and N=20. *Provenance:* The constituent-stock $\bar{\hat{\gamma}}$ values (EEM 0.041, EWJ 0.127, EWG 0.131) were computed in exploratory analysis sessions (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible from the same yfinance constituent data as the registered K experiments. Replication code available on request.

Table 4: Tail Risk Metrics: Hybrid VT vs. Buy & Hold (SPY, 2014–2026)

Metric	Buy & Hold	Hybrid VT	Improvement
ES (1%)	−4.68%	−1.35%	−71%
ES (5%)	−2.69%	−1.02%	−62%
Worst single day	−11.59%	−1.70%	−85%
Worst 5-day return	−19.81%	−4.10%	−79%
Excess kurtosis	14.71	0.46	−97%
Skewness	−0.583	−0.143	−75%
Days $ r > 2\%$	5.64%	0.00%	−100%

Notes: Hybrid VT targets $\sigma^* = 12\%$ annualised with VIX scaling ($w_t = \sigma^* / \hat{\sigma}_t \times f(\text{VIX}_t)$). ES and kurtosis metrics reflect this specification. Pure GARCH-VT (constant target, no VIX scaling) gives $\text{ES}(1\%) = -2.74\%$ and excess kurtosis = 3.76 (reconciliation check). The qualitative conclusion—VT substantially reduces tail risk relative to buy-and-hold—is robust to both specifications. *Provenance:* Tail-risk metrics in this table were computed in an exploratory analysis session (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible from SPY daily returns 2014–2026 (Yahoo Finance, `auto_adjust=False`) using the Hybrid VT specification noted above. Replication code available on request.

Table 5: Gamma-Mechanism Mapping: GJR γ Predicts VT Alpha Mechanism (Empirical Regularity 1)

Asset	γ	β^{trend}	t -stat	Mechanism
SPY	+0.211	+0.109	18.0	Trend follower
QQQ	+0.150	+0.074	17.5	Trend follower
EEM	+0.100	+0.053	14.5	Trend follower
USO	+0.050	+0.032	12.9	Mixed
BTC	+0.030	+0.007	5.3	Weak trend
TLT	+0.006	-0.006	-1.3	Variance mgmt
GLD	-0.088	-0.055	-11.8	Contrarian
Spearman $\rho(\gamma, \beta^{\text{trend}}) = 1.000$ ($p < 0.001$)				
Pearson $r = 0.993$ ($p < 0.001$)				

Provenance: The β^{trend} coefficients, t -statistics, and rank-correlation rows (ρ , r) were computed in an exploratory analysis session (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible by regressing VT weight changes on trailing 5-day returns over each asset’s full sample (Yahoo Finance, `auto_adjust=False`) using the GJR-GARCH $\hat{\gamma}$ estimates reported in the paper’s main gamma table. Replication code available on request.

Table 6: Comprehensive Null Results: Methods That Failed to Improve the GJR-GARCH

Baseline

Method	Metric	Result	Reason
<i>Prediction accuracy (10 methods)</i>			
LSTM cascade	QLIKE	DM $p=0.27$	Residuals iid
GRU cascade	QLIKE	Not significant	Same as LSTM
GARCH-LSTM	Stability	Unstable	Factor std=1.16
HAR multi-scale	QLIKE	+0.28%	Collinear
EMD	QLIKE	-0.04%	Per-window unstable
Ridge Stacking	QLIKE	-5.3%	Zeroes all features
GARCH-X (VIX)	QLIKE	0%	Scale mismatch
Expanding window	QLIKE	-4.7%	Regime contamination
FIGARCH	QLIKE	+8.7%	Long memory $\neq$ 1-step
Hurst-augmented	QLIKE	+1.28%	No OOS transfer
<i>Strategy improvement (9 methods)</i>			
VRP trading	Sharpe	Negative	Not directional
VIX term structure	Sharpe	-0.012	Redundant with ratio
Short puts	P&L	Net negative	Tail losses
Regime overlay	Sharpe	Redundant	VT already adjusts
Adaptive threshold	Sharpe	Worse	Fixed is robust
Cross-asset VaR	R^2	+0.06pp	GARCH sufficient
VIX1D	Sharpe	-0.20	30d VIX less noisy
Cond. correlation	Sharpe	$p=0.913$	Lagging indicator
Adaptive window	VaR	Identical	Student-t dominates
<i>VIX sufficient statistic tests (6 methods)</i>			
CNN Fear & Greed	Partial r	-0.063	VIX subsumes sentiment
AAII/SKEW/Credit/VVIX	Partial r	<0.03	No incremental signal
42 FRED macro series	Partial R^2	STLFSI2 16.5%	= VIX proxy
27 TW DGBAS macro	QLIKE	Import YoY +5.6%	Only exception (DM $p=0.043$)
5-indicator ensemble	Ridge	Coefficients $\rightarrow 0$	$\alpha = 100$ zeroes all
Panel GARCH-X (5 vars)	QLIKE	-8.31%	Overfitting ($p=0.022$, worse)

A Commodity Extension

A.1 Leverage Direction Across Commodities

To test the generalizability of the leverage direction taxonomy, we estimate GJR-GARCH γ for eight commodity ETFs. A clear pattern links the price mechanism to leverage direction.

Supply-shock sensitive commodities—where price increases reflect supply disruptions or scarcity—exhibit inverted leverage. Coffee (JO) shows the strongest inverted effect ($\gamma = -0.082$, 100% quarterly negative), followed by natural gas (UNG, 83% negative) and wheat (WEAT, 66% negative). These commodities share a common feature: price spikes are driven by supply-side events (weather, disease, pipeline disruptions) that simultaneously increase uncertainty.

Demand-driven commodities—where prices primarily reflect economic activity—exhibit standard leverage. Crude oil (USO, $\gamma = +0.102$, 17% negative) is the clearest example: falling prices signal weakening demand, increasing uncertainty about future conditions.

Mixed commodities show intermediate behavior: silver (SLV, 72% negative) reflects its dual role as safe-haven metal and industrial input; platinum (PPLT, 55% negative) has similar characteristics. This supply-versus-demand framework provides a deeper economic foundation for the taxonomy: the direction of asymmetric volatility reflects whether the price-driving mechanism operates through fear/scarcity (inverted) or risk/decline (standard).

A.2 Volatility Targeting for Inverted-Leverage Commodities

We test VT for the most extreme inverted-leverage commodity: coffee (JO, $\gamma = -0.082$, 100% quarterly negative). Over 2020–2023, VT improves Sharpe from 0.40 to 0.59 (+48%) and MDD from -41.7% to -13.8% . Combined with gold (+11% Sharpe) and equity results, this confirms VT effectiveness is independent of leverage direction across a broad range of asset classes and leverage intensities.

B Time-Zone Information Transmission: Supplementary Evidence

This appendix presents supplementary evidence on cross-market information transmission between U.S. and Asian equity markets. While not a main contribution of this paper, the lead-lag relationship provides independent evidence that the leverage-direction taxonomy has cross-market implications: U.S. equity volatility (driven by standard leverage) transmits to Asian markets with predictable timing.

Non-overlapping trading hours create a persistent lead-lag: SPY-to-next-day Asian market correlation is $r = 0.376$ for Taiwan, with VIX Granger-causing 0050.TW volatility ($F = 58.8$, $p < 0.0001$). A momentum strategy—hold the local index when trailing SPY return (5–10 day lookback) is positive, otherwise cash—exceeds the Harvey $t > 3.0$ threshold in six markets using c2c returns (Table 7): Hong Kong ($t = 4.12$), Australia ($t = 4.04$), Singapore ($t = 4.03$), Korea ($t = 3.83$), Taiwan ($t = 3.75$), Japan ($t = 3.69$). India and Indonesia fail (consistent with microstructure frictions), as do all U.S.-listed ETFs tracking these markets (confirming the time-zone gap mechanism) and European markets (trading-hour overlap).

Critical caveat: c2c vs. o2o returns. The c2c Sharpe ratios include overnight opening gaps that mechanically reflect U.S. price discovery. Approximately 78% of c2c alpha is not capturable. Using o2o returns, Taiwan Sharpe drops from 1.62 to 0.87 ($t = 2.22$), *failing* the Harvey threshold. The c2c results are scientifically valuable as evidence of information transmission speed but should not be interpreted as achievable returns.

Table 7: Time-Zone Momentum Strategy: Close-to-Close vs. Open-to-Open Performance

Market	c2c Sharpe	o2o Sharpe	c2c t	MDD	Sw/yr	Cost (bp)	Harvey (o2o)?
Hong Kong	1.53	—	4.12	−14.2%	38	10	—
Australia	1.50	—	4.04	−12.8%	35	15	—
Singapore	1.49	—	4.03	−13.5%	36	15	—
Korea	1.42	—	3.83	−15.1%	40	20	—
Taiwan	1.62	0.87	3.75	−9.5%	42	30	No
Japan	1.23	0.78	3.69	−16.3%	39	20	No
India	0.74	—	1.98	−22.4%	44	50	No
Indonesia	0.61	—	1.63	−25.7%	41	50	No

A global portfolio combining U.S. VT and Taiwan TZ momentum achieves c2c Sharpe 1.61 ($t = 4.31$, MDD -8.4%), but implementable o2o performance would be substantially lower. These results measure diversification potential between variance management and cross-market information channels.

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